WRF Modeling of Deep Convection and Hail for Wind Power Applications

F. LETSON AND T. J. SHEPHERD

Department of Earth and Atmospheric Sciences, Cornell University, Ithaca, New York

R. J. BARTHELMIE

Sibley School of Mechanical and Aerospace Engineering, Cornell University, Ithaca, New York

S. C. PRYOR

Department of Earth and Atmospheric Sciences, Cornell University, Ithaca, New York

(Manuscript received 25 February 2020, in final form 11 August 2020)

ABSTRACT: Deep convection and the related occurrence of hail, intense precipitation, and wind gusts represent a hazard to a range of energy infrastructure including wind turbine blades. Wind turbine blade leading-edge erosion (LEE) is caused by the impact of falling hydrometeors onto rotating wind turbine blades. It is a major source of wind turbine maintenance costs and energy losses from wind farms. In the U.S. southern Great Plains (SGP), where there is widespread wind energy development, deep convection and hail events are common, increasing the potential for precipitation-driven LEE. A 25-day Weather Research and Forecasting (WRF) Model simulation conducted at convection-permitting resolution and using a detailed microphysics scheme is carried out for the SGP to evaluate the effectiveness in modeling the wind and precipitation conditions relevant to LEE potential. WRF output for these properties is evaluated using radar observations of precipitation (including hail) and reflectivity, in situ wind speed measurements, and wind power generation. This research demonstrates some skill for the primary drivers of LEE. Wind speeds, rainfall rates, and precipitation totals show good agreement with observations. The occurrence of precipitation during power-producing wind speeds is also shown to exhibit fidelity. Hail events frequently occur during periods when wind turbines are rotating and are especially important to LEE in the SGP. The presence of hail is modeled with a mean proportion correct of 0.77 and an odds ratio of 4.55. Further research is needed to demonstrate sufficient model performance to be actionable for the wind energy industry, and there is evidence for positive model bias in cloud reflectivity.

KEYWORDS: Convective storms; Hail; Numerical weather prediction/forecasting; Renewable energy

1. Introduction

Phenomena associated with deep convective storms such as hail, downbursts, extreme rainfall rates, and lightning result in annual average economic losses in the United States in excess of $20 billion (Prein et al. 2017). Damage from hail alone results in economic losses in excess of $10 billion yr^{-1} and account for 70% of property insurance claims from severe storms (Loomis 2018). Hail from a series of thunderstorms in the Dallas–Fort Worth area of Texas during one day in 2011 resulted in losses of more than $800 million (Brown et al. 2015). Thunderstorms associated with severe hail, although less common than in the United States, are also a major weather hazard in other regions of the world including Europe (Punge and Kunz 2016). Hence, improved simulation of hazardous convective weather including hail and strong winds under both contemporary and possible future climate conditions is of great value to a range of stakeholders.

Depiction of convective storms represents a stringent challenge to regional models (Grabowski et al. 2019), but an increasing number of simulations with the Weather Research and Forecasting (WRF) Model (and other regional models) are being conducted using so-called convection-permitting grid spacings (of 1–4 km) (Feng et al. 2018; Prein et al. 2017, 2015; Scaff et al. 2020). Such simulations exhibit enhanced fidelity for convective storms relative to those conducted at lower resolution and employing cumulus parameterizations (Yang et al. 2017). Estimates of hail and/or intense rainfall produced by deep convection have also been derived using metrics of storm intensity including helicity and convective available potential energy (CAPE) (Cintineo et al. 2014; Púčik et al. 2017). A global assessment of hail risk based on these approaches and observed hail data showed substantially higher hail frequency in the central United States [including the southern Great Plains (SGP)] than in any other area of the world (Prein and Holland 2018). The SGP not only exhibits a high frequency of hail (Cintineo et al. 2012), but there is evidence that the frequency and intensity of deep convection and hail occurrence in this region has increased over the recent decades (Tang et al. 2019). The frequency of deep convection may further increase under global nonstationarity potentially resulting in an increase in hail frequency and the intensity of maximum precipitation rates, largely due to the projected increase in CAPE (Prein et al. 2017).

The SGP is also the location with the highest installed capacity of wind turbines in the United States, with as of December 2019, 27% of the 100 GW of installed capacity

Corresponding authors: F. Letson, fl368@cornell.edu; S. C. Pryor, sp2279@cornell.edu

DOI: 10.1175/JAMC-D-20-0033.1

© 2020 American Meteorological Society. For information regarding reuse of this content and general copyright information, consult the AMS Copyright Policy (www.ametsoc.org/PUBSReuseLicenses).

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

deployed in Texas and 8% in Oklahoma (American Wind Energy Association 2019). Recent research has suggested that collisions of rotating wind turbine blades with falling hydrometeors (particularly large rain droplets and hail) leads to accumulated material damage, erosion of the blade coating, and roughening of the blade leading edge (Bartolomé and Teuwen 2019; Bech et al. 2018; Letson et al. 2020b; Zhang et al. 2015).

Erosion (i.e., material loss) of the wind turbine leading edge [leading-edge erosion (LEE)] decreases aerodynamic lift and therefore is associated with reduced electrical power production (Cortés et al. 2017; Eisenberg et al. 2018; Herring et al. 2019; Traphan et al. 2018). LEE further increases operations and maintenance (O&M) costs, because of the need to repair or replace wind turbine (WT) blades (Carroll et al. 2016; Herring et al. 2019), and is the dominant cause of unplanned blade repair (Mishnaevsky and Thomsen 2020). Although leading-edge protection can be added to WT blades, such protective coverings also degrade aerodynamic performance (Herring et al. 2019). As previously reported, "In March 2018, Siemens Gamesa had to perform 'emergency' blade repair to 140 of the 175 turbines in the 630 MW London Array windfarm due to earlier than anticipated leading edge erosion. This came a month after Siemens Gamesa was forced to remove [and repair] 87 of 111 turbines in a 400 MW farm in Anholt, Denmark. In both cases, the turbines were 3.6 MW with a rotor diameter of 120 m and installed in 2013" (Herring et al. 2019, p. 2).

LEE is a topic of increasing concern to the wind energy industry and may be exacerbated by two key trends within the industry: deployments of higher capacity wind turbines with larger hub heights and rotor diameters (American Wind Energy Association 2019; International Energy Agency 2020) and increases in the projected operating lifetimes of wind turbines to 30 years (Wiser and Bolinger 2019). The speed at which tips of the WT blade rotate increases with blade length. For current and next generation commercial WT tip speeds are typically approximately 80 m s^{-1} (Dykes et al. 2014) (Fig. 1). Higher tips speeds of approximately 100 m s^{-1} may be associated with substantial decreases (of 2% to 5%) in levelized cost of energy (LCoE) (Ning and Dykes 2014). WT deployed offshore may employ higher tip speeds due to reduced noise constraints offshore in order to reduce gearbox weight and the LCoE (Dykes et al. 2014). Increased rotor tip speeds may enhance the probability of LEE offshore (Cortés et al. 2017; Herring et al. 2019), and in some locations there is evidence that coastal and offshore environments may exhibit a higher frequency of highly erosive events (Hasager et al. 2020).

The impact energy transferred to the rotating blades from falling hydrometeors is a function of the velocity of impact (i.e., closing velocity between the two) and the mass of the hydrometeor. Closing velocities are a vector sum of blade velocity (0–80 m s^{-1}), wind velocity, and the velocity of falling hydrometeors (terminal velocities 0–10 m s^{-1}). When WT blades are rapidly rotating, the kinetic energy transferred into the blades by collisions with falling hydrometeors is substantial as are the resulting materials stresses (Eisenberg et al. 2018; Fiore et al. 2015). In extreme cases, those stresses may exceed the damage threshold of composite materials from a single impact

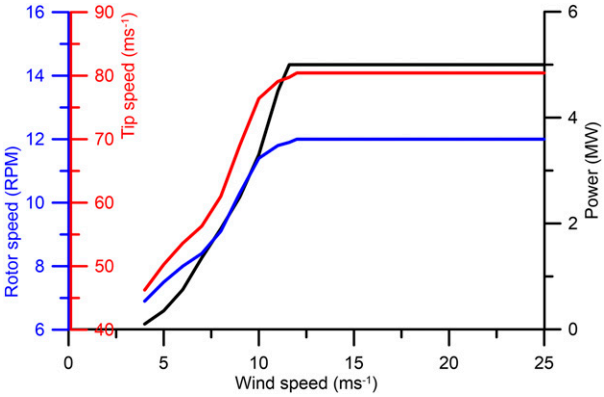

FIG. 1. The rotor rotational speed [rotations per minute (RPM)] and tip speed (m s^{-1}), along with power production (MW) as a function of wind speed for the National Renewable Energy Laboratory (NREL) 5-MW test wind turbine (Dykes et al. 2014; Jonkman et al. 2009).

(Appleby-Thomas et al. 2011; Kim and Kedward 2000; Letson et al. 2020b). The time scale to blade damage (i.e., LEE) is determined by the wind turbine blade material properties and presence of manufacturing defects (Mishnaevsky and Sütterlin 2019) and application of protective coatings as determined by the wind turbine manufacturer and wind farm developer (Mishnaevsky et al. 2017), in addition to the magnitude of environmental stressors, such as hydrometeor impacts and UV exposure. There is evidence that hail is a particularly important contributor to annual kinetic energy transfer to WT blades and thus LEE in some locations (Letson et al. 2020b; Macdonald et al. 2016). Hence, the geographic focus of this research over an area with high WT installed capacity and a high frequency of hail and extremely heavy rainfall. We focus exclusively on atmospheric drivers of LEE, specifically the hydroclimate (precipitation frequency, droplet size, and phase), wind speed regime, plus the joint probability of hail and/or heavy rainfall during periods with power-producing wind speeds. Following prior research, we infer that these atmospheric parameters are critical to dictating the LEE potential (Hasager et al. 2020).

Given the importance of LEE to the O&M costs of wind energy projects and thus the levelized cost of energy from wind power (Herring et al. 2019; Wetzel 2019), there is interest in developing methods to quantify a priori the potential likelihood and severity of precipitation-driven LEE as a function of the WT deployment location. Developing actionable estimates of LEE potential requires high fidelity in simulations of the precipitation climate including the marginal probability distributions of rainfall intensity and hail and the joint probability of heavy precipitation/hail during power-producing wind speeds over the lifetime (30 years) of the wind project. Research presented here supplements earlier radar-derived estimates of LEE potential (Letson et al. 2020b) and is part of an initial step toward developing a quantitative description of the spatial variability in LEE potential across the contiguous United States. Here we focus on evaluating the credibility of WRF simulations of the key drivers of LEE. Establishing credibility for these parameters is a necessary prerequisite to use of high-resolution

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

TABLE 1. Model domain configuration and physics settings used in the WRF simulation.

Domain settings	d01	d02	d03
	Domain settings		
Horizontal resolution (km)	12	4	1.33
Grid points (domain cells)	320 × 320	307 × 307	502 × 502
Domain extent (km)	3840 × 3840	1228 × 1228	667 × 667
Parent time step ratio	1	3	3
Time step (seconds)	72	24	8
Vertical resolution (model levels)	41	41	41
	Physics settings (all domains unless stated)		
Microphysics	Milbrant–Yau (Milbrandt and Yau 2005)		
Longwave radiation	Rapid Radiative Transfer Model (RRTM; Mlawer et al. 1997)		
Shortwave radiation	Dudhia (Dudhia 1989)		
Surface layer scheme	Monin–Obhukov similarity scheme (Jiménez et al. 2012)		
Land surface scheme	Noah land surface model (Chen and Dudhia 2001)		
Planetary boundary layer	Mellor–Yamada–Nakanishi–Niino 2.5 (Nakanishi and Niino 2006)		
Cumulus parameterization	Kain–Fritsch (Kain and Fritsch 1993)	Explicit (none)	Explicit (none)
No. of soil layers	4		
No. of land categories	21 (Noah-modified 21-category IGBP-MODIS land use)		

WRF simulations for quantifying the magnitude of, and spatial variability in, LEE potential across the nation.

2. Data and methods

a. WRF Model configuration

The WRF Model (Skamarock et al. 2008), version 3.8.1, was used for the simulation reported here. WRF is used extensively within the wind energy research and operational community for wind resource assessments and short-term forecasting (Hahmann et al. 2020; Prósper et al. 2019). Previous research indicates WRF is able to successfully reproduce organized convective systems when applied at horizontal resolutions that allow explicit simulation of convection (i.e., ≤4 km) (Kain et al. 2006; Weisman et al. 1997; Yang et al. 2017). Additional studies have shown success in simulation of inferred polarimetric signatures in supercells (Johnson et al. 2016); hail size and growth throughout a convective life cycle (Adams-Selin and Ziegler 2016); and timing, duration, and onset of hail events (Feng et al. 2018; Letson et al. 2020a).

The model is configured with three domains and uses the physics settings shown in Table 1. The model outer domain (Fig. 2) at 12 km by 12 km horizontal grid spacing covers an area of 1.47×10^7 km^2 to capture conditions at the synoptic scale. The first nested domain employs a grid resolution of 4 km by 4 km, while the high-resolution innermost nest employs a grid spacing of 1.33 km by 1.33 km and covers a geographic extent of 4.46×10^5 km^2. It is designed to capture the development and advection of convective storms over a region with high wind turbine installed capacity. The model time step is 72 s in the outer domain, and employs a step ratio of 3, such that the innermost domain time step is 8 s. A cumulus parameterization is used in the outer domain only. The vertical resolution of 41 vertical levels, with a model top of 50 hPa, is configured to place 18 model levels in the lowest 1 km of the atmosphere, to represent the planetary boundary layer (PBL) and the profile across the WT rotor plane at moderate resolution. Previous research has indicated some level of sensitivity in gross power production from the Fitch wind farm parameterization to the number of vertical levels used (Pryor et al. 2020b; Tomaszewski and Lundquist 2020). Analyses of modeled systemwide power production from all wind turbines in Iowa found relatively good agreement with observed net power production using simulations with 41 vertical levels and a 4 km grid spacing (Shepherd et al. 2020). Doubling the number of vertical levels in the lower half of the troposphere (i.e., increasing the number of levels from 41 to 57 with a 4 km horizontal resolution) led to no change in unstable conditions but increased power production by up to 5% in stable conditions (Pryor et al. 2020b). The WRF user community has recently identified a possible error within the implementation of the Fitch wind farm parameterization that may be preventing proper advection of wind turbine generated turbulence and thus artificially enhancing wind farm wake extents (Pryor et al. 2020c). This would tend to lead to an underestimation of power production and wind turbine tip speeds in the current work. This error has since been corrected in the release of WRF, version 4.2.1.

Lateral boundary conditions are supplied from ERA-Interim data (Dee et al. 2011) at 6-hourly intervals, and daily SST data for the Gulf of Mexico is provided by the Real Time Global SST (RTG-SST) dataset (Reynolds and Chelton 2010). The model is not subject to nudging, and one-way nesting is employed. Simulation output is stored every 10 min in domains 2 and 3.

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

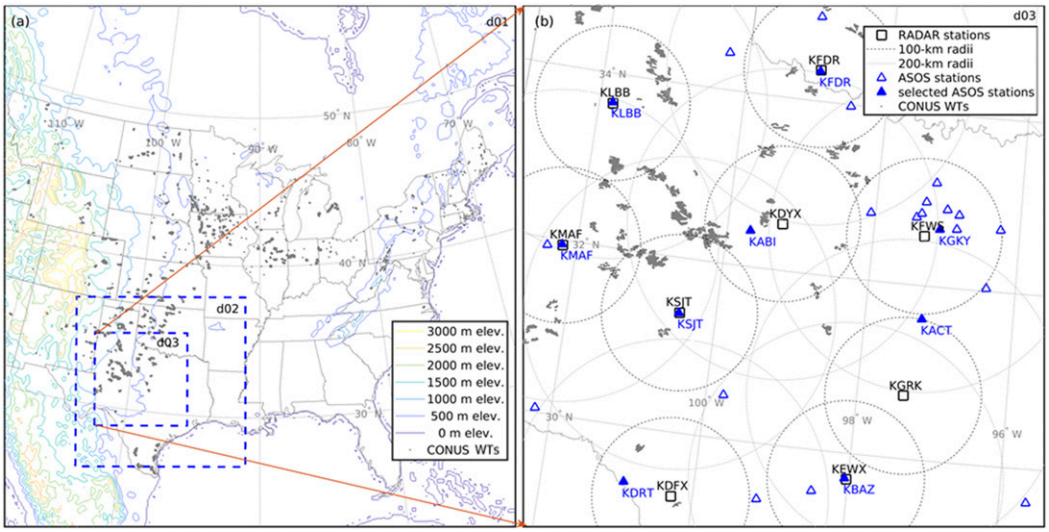

FIG. 2. (a) Topography and location of wind turbines within the three WRF simulation domains. Only wind turbines within the contiguous United States are considered herein. (b) Location of radars (black-outlined squares, and their associated 100- and 200-km radii used in the evaluation of WRF output) and NWS ASOS sites (blue triangles) within d03. The nine ASOS sites used in the WRF evaluation are indicated by the filled blue triangles.

The spatial structure and precipitation intensity as simulated by WRF exhibits a high sensitivity to the microphysics scheme (Feng et al. 2018; Morrison and Milbrandt 2011; Shpund et al. 2019; Tao et al. 2016), and inclusion of hail processes have been shown to significantly contribute to rainfall intensity, radar reflectivity, and the structure of simulated convective systems (Johnson et al. 2016; Shpund et al. 2019; Tao et al. 2016). A range of microphysics options are available within WRF. Simulations presented herein use the Milbrandt–Yau double-moment 7-class microphysics scheme (Milbrandt and Yau 2005), which treats graupel and hail as separate precipitation types. The scheme has double-moment (mass and number) description of cloud droplets, rain, ice, snow, graupel, and hail. The Milbrandt–Yau scheme includes estimated radar reflectivity as a diagnostic output variable, which is used herein as a basis for comparison to observed radar reflectivity. WRF reflectivity values are calculated as the sum of reflectivity contributions from each of several hydrometeor classes (clouds, rain, ice, snow, graupel, and hail). The hydrometeor contributions are calculated from the sixth moment of their size spectra, using Rayleigh theory (Milbrandt and Yau 2005). Convective storm diagnostics (e.g., maximum 10-m wind speed, maximum helicity in the 2–5-km height layer, maximum vertical velocity in updrafts and downdrafts below 400 hPa, mean vertical velocity in the 2–5-km layer, and maximum column graupel) are also output using the NWP diagnostics option.

Simulation settings also included use of a wind farm parameterization (Fitch 2015; Fitch et al. 2013, 2012) to compute power production from operating wind turbines. All WT deployed within the innermost domain (d03) during 2017 were georeferenced (Diffendorfer et al. 2017; Rand et al. 2020) and allocated to the appropriate WRF grid cell (Fig. 2). For use in the wind farm parameterization these WT are described using wind turbine specific hub heights, rotor diameter, power, and thrust curves as described in (Pryor et al. 2020a, 2018). The mean rated power of the 9121 WT in d03 at the end of 2017 and included in these simulations is 1.71 MW. The mean hub height is 79 m (standard deviation of 12 m) while the mean and standard deviation of rotor diameter are 86 and 18 m, respectively. WT deployed in the United States during 2019 had an average hub height of 90 m and a rotor diameter of 121 m, which are both marked increases from comparable values in 2015 of 82 and 102 m, respectively (American Wind Energy Association 2020).

The WRF simulation was performed for 8 June to 2 July 2014, inclusive. This 25-day period was chosen to represent summer conditions, when hail in the SGP is most frequent (Fig. 3). This period was also chosen because some radar hail

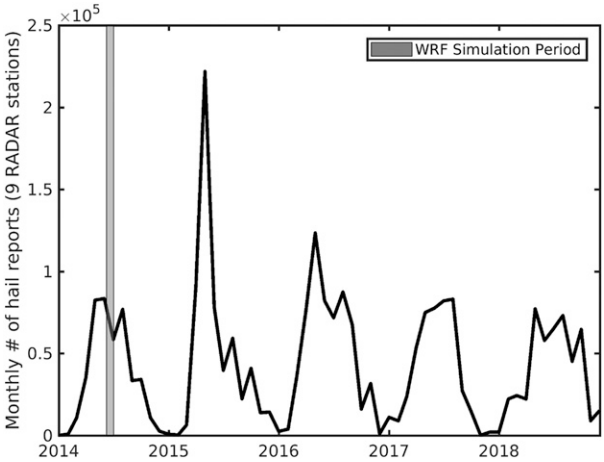

FIG. 3. Time series of monthly number of 5-min hail reports, totaled across the nine radar stations in the current study. The duration of the WRF simulation is shown in gray.

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

reports therein have been confirmed by human observation (NOAA 2019). The simulation was performed on eight half-packed 32-core nodes on a Cray XC40 [National Energy Research Scientific Computing Center (NERSC) Cori system]. The run time was 67.3 h for a total of 8614 core hours.

b. Observations for model evaluation

1) PRECIPITATION CLIMATE

The precipitation climate as simulated by WRF is evaluated using concurrent observations from nine National Weather Service (NWS) dual-polarization radar stations (Crum et al. 1998; Seo et al. 2015) (Fig. 2b). This network consists of 159 S-band WSR-88D stations covering the majority of the United States. This network underwent an upgrade to dual polarization, which was completed in 2013, to permit identification of precipitation phase, and thus accurately distinguish between hail and heavy rain events (Crum et al. 1998; Seo et al. 2015). The high-density radar network in the SGP provides an opportunity to evaluate the WRF Model with objective hail observations. These NWS radar record atmospheric conditions within 200 to 300 km of the radar locations, at 9 elevation angles (from 0.5° to 19.5°) during typical operation, increasing to 14 elevation angles during severe weather (Office of the Federal Coordinator for Meteorology 2017). The azimuth resolution is 1°. Depending on precipitation conditions in the scanned volume, scan frequency varies from 5 (storm mode) to 10 (clear-air mode) min. The following radar products are used at a 10-min resolution in the WRF evaluation:

1) Composite radar reflectivity in decibels (dBZ) (“NCR”; the maximum reflectivity in each vertical column) is used.
2) Precipitation rate [one-hour precipitation (N1P)] in millimeters per hour derived using Z–R relationships (Office of the Federal Coordinator for Meteorology 2016) is used.
3) Hail reports [hail index (NHI)], issued on the basis of reflectivity, aspect ratio of hydrometeors, vertically integrated liquid, and altitude of the melting layer (Office of the Federal Coordinator for Meteorology 2017) are used. Radar algorithms for detection of hail have been experimentally evaluated (Wallace et al. 2019) and shown to be especially effective over the SGP (Witt et al. 1998). Hail reports are issued every 5 min during periods when hail-producing convective storms are present and include the geographic position and the 75th percentile hailstone diameter [maximum estimated size of hail (MESH)] for each such storm.

The wind turbine installed capacity varies greatly across domain 3 (Fig. 2a) and in the volume scanned by the nine radars (Table 2). Installed wind turbine capacities of 2833 and 2360 MW exist within 100 km of two radar stations in the northern half of domain 3 (KDYX and KLBB). These locations are thus of particular interest for projections of wind turbine LEE, and hence, WRF performance at these locations receives particular attention in the following.

2) WIND CLIMATE

NWS radar stations also provide estimates of radial wind speeds via the Doppler shift (Alpert and Kumar 2007), at heights of relevance to wind turbine rotor planes, where typical hub heights of 80–100 m are scanned by the lowest elevation angle (0.5°) at approximately 8-km radius from the radar station (Letson et al. 2020b). Highest fidelity is achieved for homogeneous wind fields with high wind speeds, with lower skill for low–moderate winds (Forget et al. 2016). Thus, WRF wind speeds at 10-m AGL are evaluated relative to observations from nine two-dimensional sonic anemometers of the NWS Automated Surface Observing System (ASOS) (Schmitt and Chester 2009). A key issue with evaluating near-surface wind speeds output from a numerical model relative to point in situ observations pertains to the mismatch in spatial aggregation and the influence on the measurements from nearby obstacles and surface roughness inhomogeneities (Masters et al. 2010; Pryor and Hahmann 2019). Visual inspection of orthophotographs of each ASOS station in domain 3 were used to select nine ASOS stations for use in the WRF evaluation (Fig. 2b) that are surrounded by open terrain and few nearby buildings to obstruct wind flow. ASOS wind observations are made at 10 m AGL and recorded at a resolution of 1 kt (1 kt $\approx$ 0.51 m s^{-1}; rounded *up* to the nearest 0.514 m s^{-1}). Wind speeds below 3 kt (1.543 m s^{-1}) are recorded as “calm” or 0 m s^{-1} (NOAA 2004). The 5-min ASOS wind speeds observations are converted to 10-min mean values for direct comparison with the WRF output.

Wind speeds at which wind turbines rotate and produce electrical power (Fig. 1) are of particular importance to wind turbine LEE. Thus, wind turbine electrical power production estimates from the Fitch wind farm parameterization in the WRF simulations are also evaluated relative to observed wind energy power production in each 15 min period as aggregated across the Electric Reliability Council of Texas (ERCOT) power system (Fertig 2019). Since domain 3 is not perfectly congruent with the ERCOT region (which covers most but not all counties in Texas), this comparison is conducted using normalized power, wherein power from each 15 min period (from ERCOT or WRF, linearly interpolated from the two

TABLE 2. Radar station identifiers and locations. Also shown is the installed capacity (IC) of WT in megawatts within 100 and 200 km of each radar. The WT IC is derived on the basis of data from the U.S. Geological Survey (USGS) database and reflects the dataset used in the WRF simulations presented herein (Diffendorfer et al. 2017).

Radar code	Lat (°N)	Lon (°W)	Wind power capacity within 100 km (MW)	Wind power capacity within 200 km (MW)
KDFX	29.273	100.281	248	248
KDYX	32.538	99.254	2833	8813
KEWX	29.704	98.028	0	149
KFDR	34.362	98.976	1111	3935
KFWS	32.573	97.303	6	1799
KGRK	30.722	97.383	0	349
KLBB	33.654	101.814	2360	6737
KMAF	31.943	102.189	1895	8297
KSJT	31.371	100.493	1792	9150

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

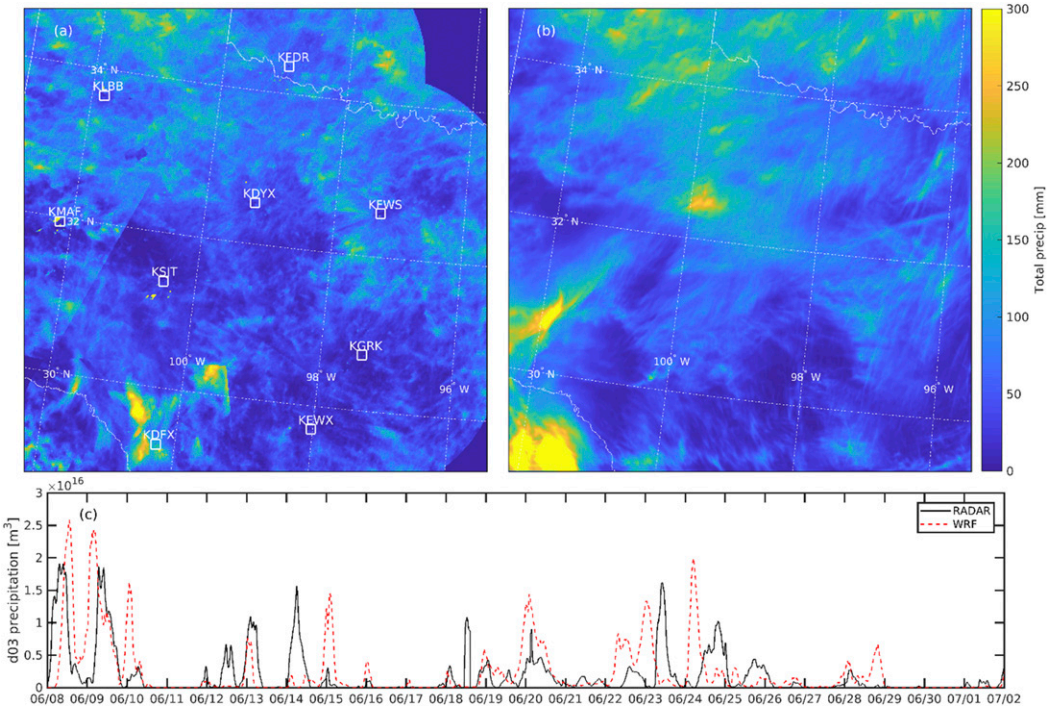

FIG. 4. Precipitation totals for the 25-day simulation period from (a) radar and (b) WRF domain 3 (1.33 km). The radar precipitation data are mapped onto the WRF grid. (c) Time series of 10-min domainwide total precipitation volume from WRF and radar. The discontinuity in precipitation totals shown in (a) between KMAF and KSIT is explained by the use of different Z–R relationships for the calculation of precipitation rates around each radar station.

adjoining 10-min periods) is normalized to the maximum value during the 25 day period.

c. Model evaluation methods

The purpose of this analysis is to provide a first assessment of the fidelity of WRF with respect to the atmospheric drivers of WT LEE. As such we focus on evaluation of the accuracy of rain, hail, derived radar reflectivity, wind speeds, and wind power output from the Fitch wind farm parameterization in WRF relative to radar and in situ observations.

Spatial variability in total precipitation from the WRF output is directly compared to radar-derived estimates mapped onto the 1.33 km by 1.33 km (domain 3) WRF grid. Evaluation and verification of hail occurrence and amount is more challenging. There is particular value in, but challenges to, verification of severe hail (Wendt et al. 2016). Hail observations from radar are point data indicating likely hail occurrence with no corresponding spatial extent or hail accumulation estimate. Thus, we compare the total hail accumulation in WRF grid cell to the number of radar hail reports in each grid cell over the 25-day period. Time series comparisons for rain and hail are also made to evaluate the accuracy of the timing of precipitation events in the WRF output. In these analyses, the total volume of precipitation accumulated in all WRF grid cells is compared to the domainwide total radar-estimated precipitation. The spatial extent of hail is evaluated for each 10-min period by comparing the total number of WRF grid cells with nonzero hail accumulation with the number of radar-derived hail reports summed over all WRF grid cells. Evaluation of WRF performance is also assessed for each individual radar location using two distance thresholds from the radar, 100 and 200 km.

Similar geospatial and temporal analyses are undertaken for composite reflectivity estimates from WRF and radar. Two reflectivity thresholds provide a focus for these analyses: 40 dBZ, above which hail may be occurring, and 50 dBZ, above which hail is extremely likely (Witt et al. 1998).

This research is designed as a first step toward evaluating the potential for use of WRF in describing the atmospheric drivers of WT blade LEE potential, it is important to evaluate the marginal and joint probability distributions of wind, rain, and hail events. Hence cumulative density functions (CDFs) of rainfall rate, reflectivity, and wind speed are evaluated along with the probability distributions of wind speed conditioned on the occurrence of precipitation. For rainfall rate and wind speeds, spatiotemporal CDF values are calculated by finding the fraction of radar or WRF grid cells over the simulation period that exceed a given threshold (calculations are made for radii of 100 and 200 km around each radar station). Wind speed CDFs are also calculated using each of nine ASOS time series (averaged to a 10-min interval) and the 10-m WRF wind speed from the WRF grid cell containing the ASOS station.

Occurrence of hail is of particular importance to LEE and is evaluated using a contingency-table-based approach. In this analysis comparisons are made for the occurrence of hail within 100 km of each radar station at an hourly increment to avoid double counting of hailstorms that have a duration

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

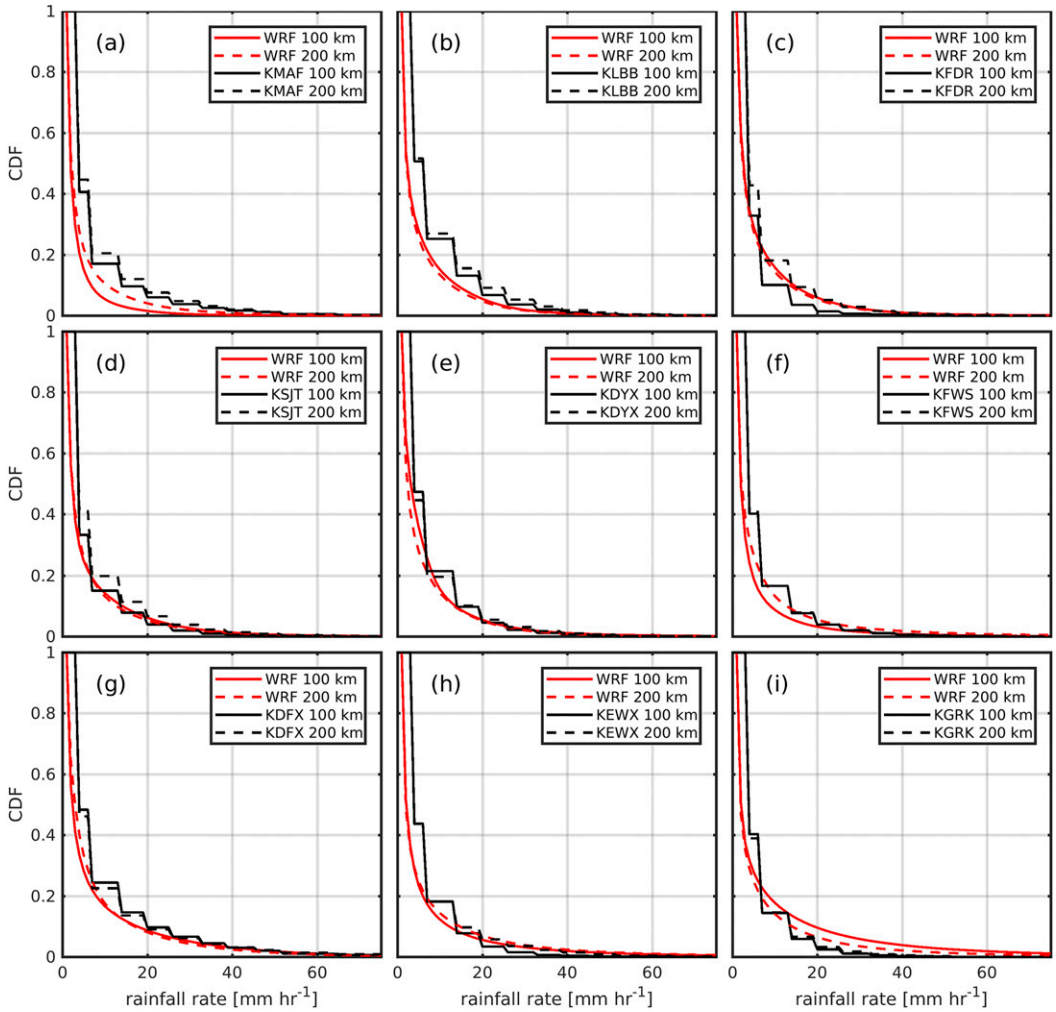

FIG. 5. CDFs of 10-min rainfall rates from WRF and each radar station for areas within 100 and 200 km of each radar station. Note that radar rainfall rates are reported using a categorical scale rather than as a continuous variable.

longer than the 5-min radar sampling intervals. Comparisons are made hourly for the presence or absence of hail, and for intense hail events: defined as hours with a number of hail reports above the median value (for hours with one of more hail reports) in the radar data, and total hail volume above the median value (for hours with $>0\,\mathrm{m}^3$ hail accumulation) in the WRF output. The statistical metrics reported are

$$\text{Proportion correct: } C = \frac{a+d}{a+b+c+d}, \tag{1}$$

$$\text{Hit rate: } H = \frac{a}{a+c}, \tag{2}$$

$$\text{False-alarm rate: } F = \frac{b}{b+d}, \text{ and} \tag{3}$$

$$\text{Odds ratio: } \theta = \frac{H}{1-H}\bigg/\frac{F}{1-F}, \tag{4}$$

where a, b, c, and d are the numbers of hits (correct predictions of an event), false alarms (predictions of an event when none occurred), misses (event occurred without a prediction), and correct negatives (correct forecast of no event), respectively.

Spearman (rank) correlation coefficients ρ are also computed for the time series of WRF wind speed and power from the Fitch scheme relative to ASOS and ERCOT wind power generation, respectively. A permutation test is used to compute confidence intervals to evaluate the corresponding correlation coefficients (Wilks 2011). In this analysis 100 000 iterations are drawn of random number time series are derived that have the same effective sample size as the wind speed ($N \sim 3300$) or power time series and used to compute a sample of correlation coefficients against which the observed ρ can be compared. The observed ρ values are different from zero at a given significance level p if they lie beyond the $1-p$ tails of the distribution of ρ computed from the random number draws.

3. Results and discussion

WRF simulated rainfall on the 1.33-km domain exhibits similar magnitude and spatial variability to radar observations for the study period, although the location and timing of heavy

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

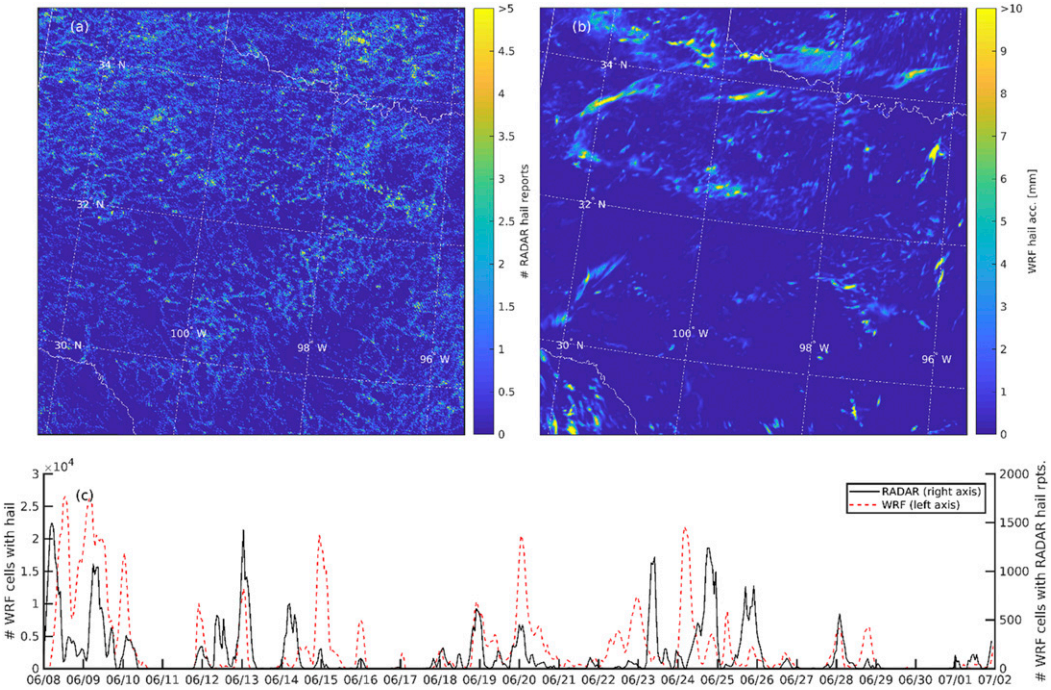

FIG. 6. (a) Total number of radar hail reports in each WRF 1.3 by 1.3 km grid cell. (b) Total hail accumulation (mm) in each WRF grid cell during the entire 25-day simulation period. (c) Time series of the number of WRF cells in each hour with nonzero hail accumulation and the number of WRF cells for which the radar data imply presence of hail.

rain events is not in perfect agreement with radar (Fig. 4). Radar-derived estimates of total precipitation exhibit higher geospatial variability, and specifically smaller scales of coherence, especially in the north of the domain (Figs. 4a,b). Domainwide rainfall totals exhibit similar temporal structure (Fig. 4c). Comparison of spatiotemporal rainfall rate CDFs near each radar station shows no consistent bias in the WRF output. The station-to-station variability in rainfall rates aggregated over both the 100- and 200-km scales is well modeled by WRF (Fig. 5). For example, rainfall rates $> 20\,\mathrm{mm\,h^{-1}}$ are more common near KDFX (Fig. 5g) than in other areas of domain 3, and this is captured in the WRF output. The precipitation CDFs for KLBB and KDYX (Figs. 5b,e; the areas of highest WT concentration in domain 3; Table 2), show similar or better fidelity with respect to radar observations than at the other radar stations (Fig. 5). A further implication of Fig. 5 is that similar model performance is observed for the two levels of spatial aggregation (i.e., radii of 100 and 200 km).

WRF simulated hail accumulation exhibits greater spatial coherence than the radar-derived hail reports (Fig. 6). An autocorrelation analysis of these two spatial fields (Wilks 2011) shows that WRF hail accumulation totals have significant spatial correlation at scales of up to ~30 km, whereas RADAR hail reports show very little correlation beyond ~7 km. This discrepancy exists in part because a single hail report is issued for each storm cell, while the area of hail may extend over multiple WRF grid cells. The duration of widespread hail (10-min periods in which >5000 WRF cells show hail accumulation in the WRF output, or >250 WRF cells contain radar hail reports) is in good agreement (they tend to be 12–24 h in duration; Fig. 6c). The number of WRF cells with hail in each 10-min period is approximately 15 times the number of WRF cells that contain radar hail reports. This implies that radar hail reports may correspond to hail events that cover ~15 WRF cells (~35 km^2, or a radius of 3.4 km). This is consistent with the size of hail-producing areas of thunderstorms observed in the southern United States (Buechler and Goodman 1990; Stull 2017).

Statistics derived from contingency tables for hail events show a proportion correct C of 0.69–0.81 for hail-containing hours within 100 km of each radar station (Table 3). This increases to 0.83–0.91 when only hail events of above median intensity (above the median number of radar hail reports or WRF cells with hail accumulation) are considered. Odds ratios θ range from 0.89 to 10.46 when all hail-containing hours are considered and from 1.06 to 25.22 when only hours above median intensity are considered. Recall, $\theta > 1$ are an indicator of better-than-random model performance (Stephenson 2000). This means that when WRF output indicates the presence of hail within 100 km of a radar station, the odds are (on average) 4.5 to 1 that radar would indicate the presence of hail. If WRF output shows a hail hour of intense hail odds are 7.7 to 1 that radar report intense hail for the same period. Proportion correct and odds ratio in the areas of highest WT concentration (near KLBB and KDYX) are within the range of values calculated for the other nine stations, meaning that WRF performance at modeling hail occurrence is not significantly different in locations within the simulation domain that exhibit widespread WT deployment.

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

TABLE 3. Contingency tables comparing WRF hail events with radar observations, for hours with any hail activity (top half of table) or hours with hail activity above median intensity (bottom half of table). Each column shows the number of hits (a), number of false alarms (b), number of misses (c), number of correct negatives (d), proportion correct (C), hit rate (H), false-alarm rate (F), and odds ratio (θ).

	KDFX	KDYX	KEWX	KFDR	KFWS	KGRK	KLBB	KMAF	KSJT
				All hail events					
a	25	21	23	32	45	39	62	40	8
b	78	86	82	135	85	69	146	105	97
c	29	38	26	32	32	24	30	54	40
d	444	431	445	377	414	444	338	377	431
C	0.81	0.78	0.81	0.71	0.80	0.84	0.69	0.72	0.76
H	0.46	0.36	0.47	0.50	0.58	0.62	0.67	0.43	0.17
F	0.15	0.17	0.16	0.26	0.17	0.13	0.30	0.22	0.18
θ	4.91	2.77	4.80	2.79	6.85	10.46	4.78	2.66	0.89
				Events above median intensity					
a	11	7	11	9	15	19	21	17	2
b	40	46	41	74	50	35	83	55	50
c	15	22	12	23	23	11	23	30	19
d	510	501	512	470	488	511	449	474	505
C	0.90	0.88	0.91	0.83	0.87	0.92	0.82	0.85	0.88
H	0.42	0.24	0.48	0.28	0.39	0.63	0.48	0.36	0.10
F	0.07	0.08	0.07	0.14	0.09	0.06	0.16	0.10	0.09
θ	9.35	3.47	11.45	2.49	6.37	25.22	4.94	4.88	1.06

Kinetic energy transferred from falling hydrometeors to WT blades is a function of the closing velocity (a vector addition of the terminal fall velocity of the hydrometeor, the rotational velocity of the WT blade, and the wind speed) and hydrometeor phase and size. Previous research has indicated the importance of severe (large) hail to accumulated materials stress (Bartolomé and Teuwen 2019; Bech et al. 2018). Thus, it is noteworthy that the radar-derived estimates of 75th

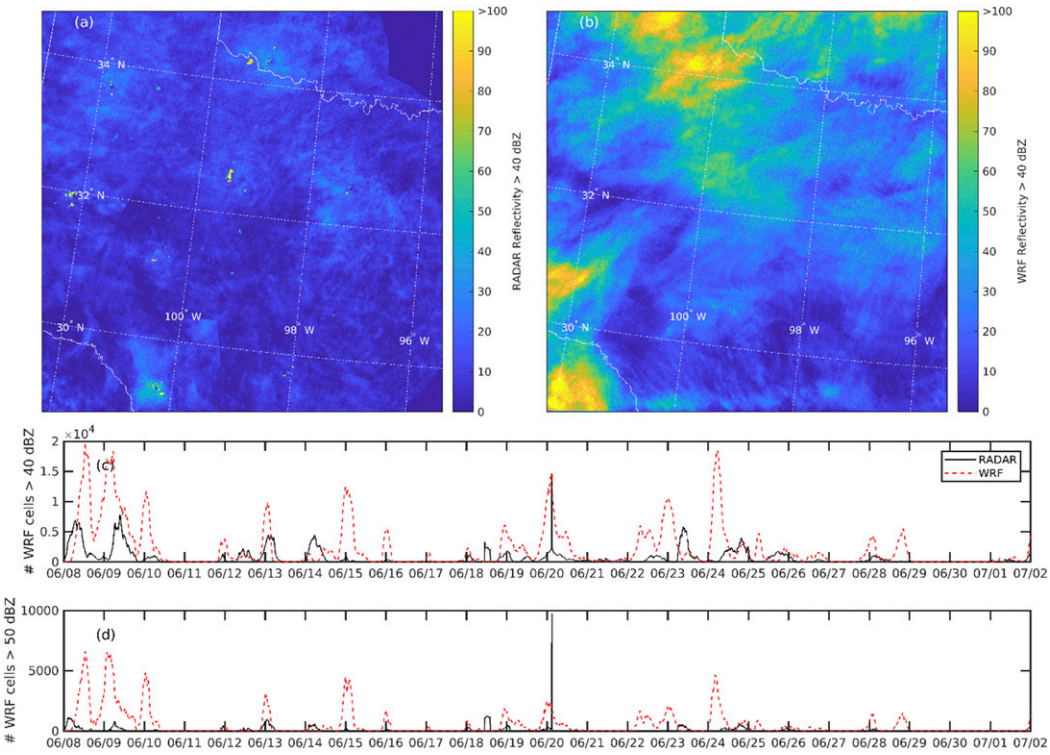

FIG. 7. Spatial maps of the number of 10-min periods for which (a) observed composite radar reflectivity exceeds 40 dBZ and (b) simulated composite reflectivity from the WRF Model output exceeds 40 dBZ in each WRF 1.3 km by 1.3 km grid cell. Also shown are time series of the number of WRF grid cells with composite reflectivities from WRF and the radar above (c) 40 and (d) 50 dBZ.

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

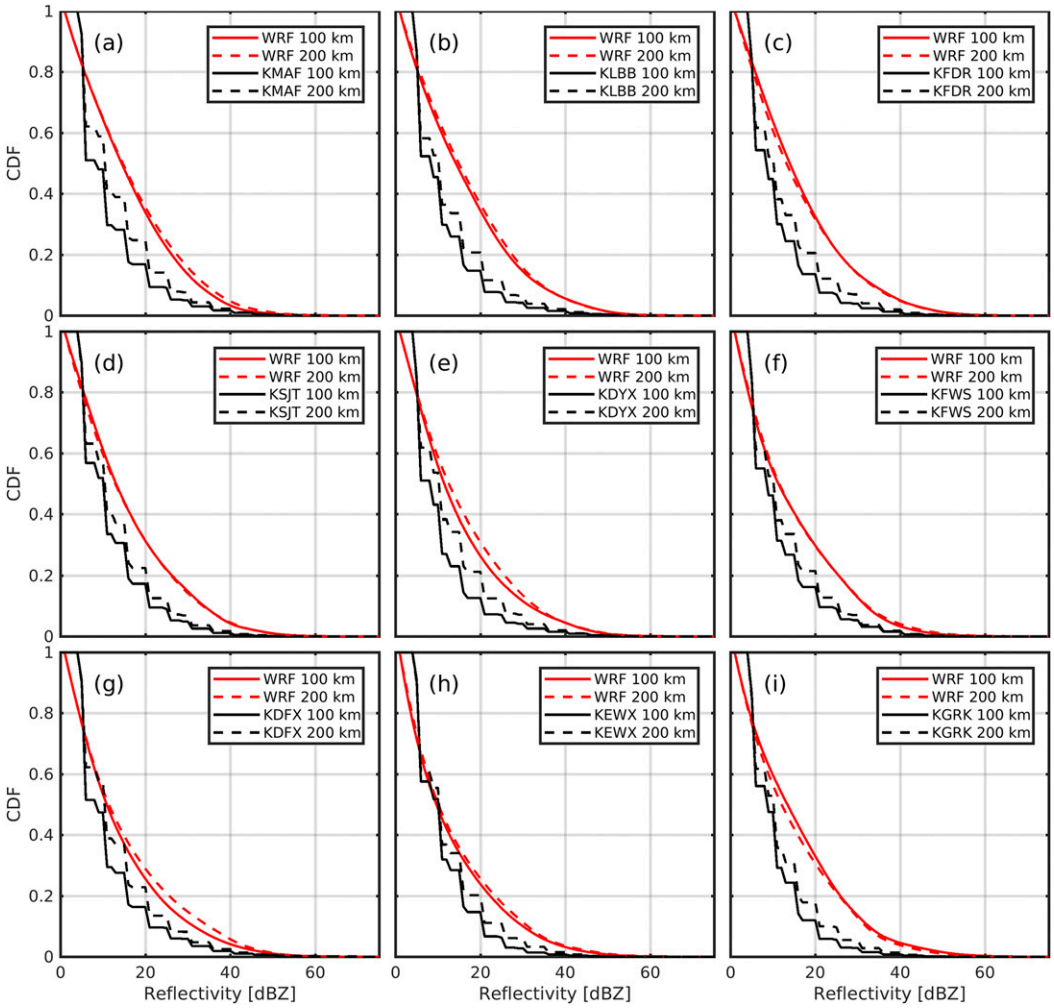

FIG. 8. CDF of composite reflectivity [p (reflectivity) $\geq X$] sampled across space and time from WRF and radar within 100 and 200 km of each radar station. Reflectivities are sampled every 10 min. The CDFs exhibit slightly different forms because of the categorical nature of the wind speed observations.

percentile hailstone diameter, D_{75} exhibit a strong correlation with the number of hail reports observed within 100 km of each radar station. The median D_{75} in a given 10-min period value increases by an average of 0.16 mm for each additional hail report in the 100-km radar station radius. It increases from ~5 mm when the number of reports is near zero to >20 mm when there are >100 concurrent reports (e.g., during periods when hail is widespread, hailstones also tend to be larger). Thus, the high fidelity of WRF for intense hail events is of particular value to accurate estimates of LEE potential.

Consistent with the majority of past research (Grabowski et al. 2019; Shpund et al. 2019), and a study that applied WRF at 1-km horizontal grid spacing with eight different microphysics schemes (including one-moment, two-moment, and hybrid approaches) (Fan et al. 2017), simulations presented herein are positively biased in terms of the areal extent of radar reflectivity in excess of 40 and 50 dBZ (Figs. 7, 8). For example, more than 60 time steps exhibit reflectivities >40 dBZ in many WRF grid cells, but when the radar data are remapped to the WRF resolution, very few grid cells exhibit more than 40 time periods with values in excess of this threshold (Figs. 7a,b). Over time, the number of WRF cells with WRF reflectivity >40 dBZ varies between 0 and 20 000 (i.e., up to 8% of domain 3), whereas the number of cells with radar reflectivity >40 dBZ is rarely more than 6000 (Fig. 7c). This difference is even more pronounced when the threshold is increased to 50 dBZ (above which hail is very likely present; Fig. 7d). The CDF of reflectivity from WRF and the individual radars reemphasizes this positive bias. The probability of a cell exhibiting reflectivity >20 dBZ is approximately 2 times as high in the WRF output as in radar observations. Reflectivity estimates (and their bias with respect to radar observations) in areas of widespread WT deployment (KDYX and KLBB; Figs. 8b,e) are similar to other areas (Fig. 8). While reflectivity is a diagnostic variable from WRF and thus does not per se lead to biases in the hydroclimate, it is likely symptomatic of errors in the representation of cloud microphysics (the size and number of hydrometeors), which may lead to biases in rainfall

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

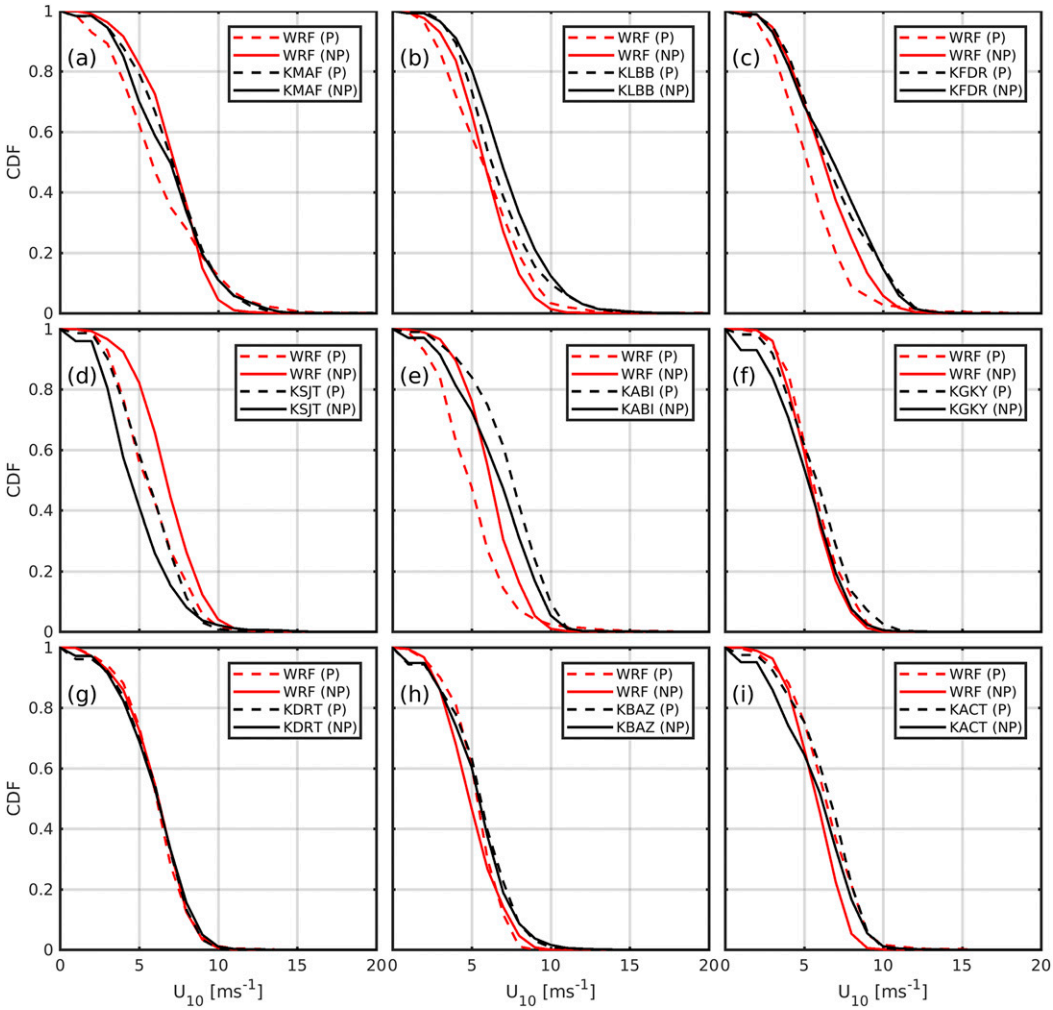

FIG. 9. CDF of wind speeds at 10-m AGL from ASOS two-dimensional sonic anemometers and from WRF for the grid cell containing each ASOS station, conditioned based on the occurrence of precipitation within 100 km of the closest radar station; P indicates precipitation is observed or simulated, and NP denotes an absence of precipitation. ASOS wind speeds are conditionally sampled for the presence of precipitation from the radar, whereas WRF wind speeds are conditionally sampled by the presence of precipitation in the WRF simulations.

rates and hail occurrence (Tao et al. 2016). This implies tremendous value in systematic assessments of different microphysics schemes across a range of convective environments. Also, the availability of validation data for hail occurrence is extremely important to advancing evaluating and making improvements to WRF simulations (Snook et al. 2016).

WRF simulation of wind climates has been extensively evaluated within the New European Wind Atlas and other research projects (Hahmann et al. 2020; Hawbecker et al. 2017; Pryor and Hahmann 2019). When applied at resolutions of 3–12 km, closest agreement has been reported over flat terrain with low surface roughness and during unstable (convective) conditions (Pryor and Hahmann 2019). These conditions are largely realized over the simulation domain, and accordingly wind speeds at 10 m from WRF exhibit good but spatially varying agreement with ASOS observations. The agreement is less good in higher-elevation areas in the northwest of the domain (Fig. 9, Table 4). In contrast to an earlier study over complex terrain (Carvalho et al. 2012), there is no evidence for substantial negative bias in the modeled wind speeds. Agreement is observed across the entire probability distribution, which is important to quantification of kinetic energy transfer to the blades from hydrometeors since blade rotational speeds vary substantially with wind speed (Fig. 1). Spatial variability in model performance is evident from the Spearman rank correlation coefficients between WRF and ASOS 10-m wind speeds at the different sites (Table 4). These correlation coefficients vary between 0.21 and 0.58 and are statistically different from 0 at the 99.9% confidence level. There is no consistent difference in wind speed distributions or in WRF Model performance (degree of agreement with ASOS) for periods with precipitation that account for 16%–28% of the ~3200 ten-min periods in the simulation and) that have highest relevance for LEE. While the degree of agreement between

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

TABLE 4. Spearman (rank) correlation coefficient ρ_s for 10-min time series of ASOS 10-m AGL wind speeds and WRF 10-m wind speeds at the nine ASOS sites; N denotes the sample size.

ASOS code	Nearest radar station	N	ρ_s
KDRT	KDFX	3273	0.2105
KABI	KDYX	3288	0.5065
KBAZ	KEWX	3211	0.5796
KFDR	KFDR	3231	0.5768
KGKY	KFWS	3168	0.4592
KACT	KGRK	3268	0.5528
KLBB	KLBB	3328	0.2818
KMAF	KMAF	3334	0.2825
KSJT	KSJT	3291	0.2752

10-m wind speeds during precipitation is good at KLBB (Fig. 9b), one of the locations with high WT installed capacities, it is least good at KABI (Fig. 9e) near radar station KDYX), in another area of concentrated WT deployment.

Wind power output totals from domain 3 are in broad agreement with historical wind power output data form ERCOT (Fig. 10b). There are two days (29 and 30 June) near the end of the simulation period for which WRF wind power output differs substantially from ERCOT observations (Fig. 10b). These periods were not associated with precipitation (Fig. 4) and thus would not have contributed to errors in LEE estimates. Wind speed profiles from WRF for these periods indicate relatively low wind speeds at WT rotor heights, that are at, or close to, the cut-in for the majority of WT deployed in Texas (4 and 7 m s^{-1}; Fig. 10a), while the ERCOT power production data during this period suggest higher wind speeds with substantial electrical power production (Fig. 10b). The overall Spearman correlation coefficient for normalized wind power output series from WRF and ERCOT is 0.52, and this increases to 0.63 when times after 28 June are excluded from the comparison. These rank correlations differ from zero at the 99.87% and 99.97% confidence levels.

The modal wind speed close to typical WT hub heights (~91 m) during hail events in WT-containing WRF cells is 8 m s^{-1}. Wind speeds vary across the rotor disk, but the modal value is the same for heights around 55 and 127 m (Fig. 10a), which covers much of the likely span of rotor disk heights for WT deployed over the next decade. This wind speed is in the steepest part of typical WT power and RPM curves (Figs. 1, 10a) and thus indicates a high sensitivity of model-derived kinetic energy transfer to the accuracy of modeled wind speeds. Kinetic energy transfer to WT blades from falling hydrometeors will likely be maximized during periods when wind speeds are in the range of 12 and 25 m s^{-1} and thus WT are operating at or close to rated power and tip speeds are maximized. Hub-height wind speeds are in this range during 35% of hail events.

4. Concluding remarks

This analysis is designed as a first step toward generation of geospatially explicit estimates of wind turbine blade leading-edge erosion potential using simulations with the WRF Model. The evaluation of WRF is conducted relative to radar precipitation and reflectivity data, ASOS wind speed observations, and wind power output records from ERCOT. Precipitation totals and rainfall rates from WRF are similar in magnitude and distribution to the radar record and show a high degree of spatial and temporal variability consistent to radar. Evaluation of hail is confounded by the nature of the observations from radar, but WRF exhibits some level of fidelity with respect to modeling the occurrence and intensity of hail, although it is positively biased in terms of frequency and spatial extent. The majority of hail events at WT locations are associated with wind speeds at which a typical wind turbine will be rotating, and a substantial fraction occur when WT will be operating at rated capacity and exhibiting highest tip speeds (Fig. 10a). Hours with simulated hail occurrence in 100-km radii around each of the nine radar stations exhibit a mean proportion correct of 0.77 (range of 0.69–0.84) and the false-alarm ratio ranges from 0.06 to 0.16. The mean odds ratio is 4.55 (range of 0.89–10.46) for the occurrence of hail relative to radar observations. The odds ratio is even higher for the most intense hail events. Wind speeds and wind power output are in good agreement with observations both in terms of the CDF at point locations and the time evolution, with the latter showing rank correlation coefficients that differ from zero at the 99% confidence level. The model also exhibits some skill with respect to the joint occurrence of precipitation and power-producing wind speeds.

In principle, a priori geospatially explicit estimates of LEE potential would be useful to the industry in multiple ways. First, they would permit wind farm developers to conduct more accurate cost–benefit analyses regarding the relative advantage of applying blade leading-edge protection at a given site (Herring et al. 2019). Second, they would provide a priori information to be used for guidance regarding planning for blade inspections and maintenance scheduling/contracts at specific locations (Nielsen et al. 2020; Papaelias and Márquez 2020). Third, they would provide critical information regarding the potential wind turbine lifetime gains that may derive from dynamic operation of the wind farm to enact nowcast curtailment of wind farms during highly erosive events (Tilg et al. 2020). The relatively short duration of hail events places a premium on prediction accuracy in both space and time, but there is the potential for significant LCoE benefits (Bech et al. 2018).

Considerable further research is needed to demonstrate that model fidelity is sufficient to be actionable for the wind energy industry. The current study concentrates on a particularly hail-prone location and season in order to maximize the number of hail occurrences available for comparison to radar observation in a limited time period. Evaluation of model performance in other environments, including offshore regions with substantial wind resources, would be a useful addition. A comprehensive assessment of model sensitivity to the specific physics options chosen would also be valuable as would demonstration of the ability to accurately model the absence of hail in a less hail-prone place and time. Long-term simulations over a larger area are necessary to accurately characterize geospatial variability in the atmospheric drivers of WT blade LEE including extreme joint wind–precipitation events (i.e., upper tails of

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

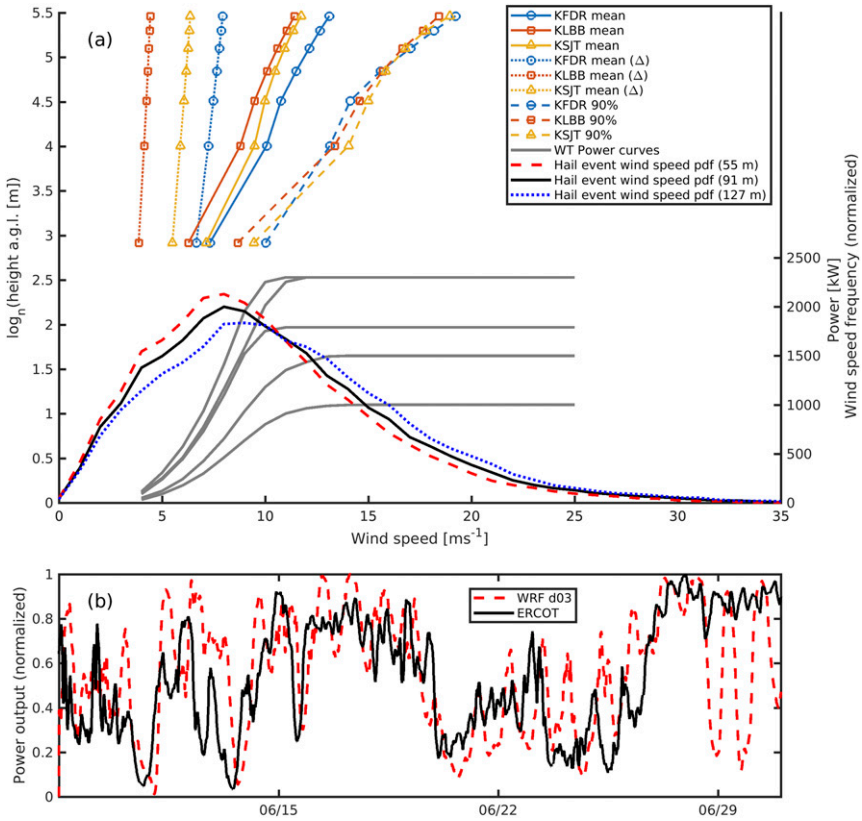

FIG. 10. (a) Wind speed profiles [mean (solid) and 90th-percentile (dashed)] output from WRF grid cells containing three ASOS stations, and power curves for the five most common WT types in domain 3 (accounting for 60% of total WTs; see details in Pryor et al. 2018). Normalized wind speed probability distributions at WT locations for all hours of hail accumulation are shown for three heights within the typical WT rotor disk. The distribution for ~91 m AGL (closest to the mean WT hub height in d03) is shown by the black line, the ~55 m AGL distribution is shown in dashed red, and the ~127 m AGL distribution is shown in dotted blue. Also shown are wind speed profiles at times when there is a >50% discrepancy between estimated power output from the wind farm parameterization employed in WRF and observed power from the ERCOT network (Δ). (b) Time series of normalized total WRF wind power output in domain 3 and observed ERCOT wind power output (normalized by the maximum value in each series). These time series are normalized to account for differences in the WT fleet in the ERCOT dataset and that located within d03.

joint probability distributions of wind and precipitation) and would require substantial computational resources. Finally, access to high-quality industry data of wind speed profiles to heights across the WT rotor-swept area and on the occurrence, characteristics, and spatial variability of LEE is essential to ground-truthing any estimates of LEE probability derived from atmospheric modeling.

Acknowledgments. This research is supported by the U.S. Department of Energy (DE-SC0016438 and DE-SC0016605). Computational resources are provided by the NSF Extreme Science and Engineering Discovery Environment (XSEDE) (Award TG-ATM170024) and the National Energy Research Scientific Computing Center, a Department of Energy Office of Science User Facility supported by that office under Contract DE-AC02-05 CH11231. The thoughtful comments of the editor and four external reviewers are gratefully acknowledged.

Data availability statement. ERA-Interim output is available from the European Centre for Medium-Range Weather Forecasts (http://apps.ecmwf.int/datasets/). The WT locations as reported by the USGS are available online (https://cmerwebmap.cr.usgs.gov/catalog/item/530796abe4b0e530a6b4eb8c). The NOAA/NCEP real-time global sea surface temperature analyses are available online (https://www.nco.ncep.noaa.gov/pmb/products/sst/). The WRF Model output generated within this project (including the namelist used) have been archived online (http://portal.nersc.gov/archive/home/projects/m2645/www/public_data_2014_tx_hail_study). NEXRAD radar data, including all products used in the current study are available from the National Centers for Environmental Information

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

(https://www.ncdc.noaa.gov/data-access/radar-data). ERCOT power generation data (including wind power output) are available from ERCOT (http://www.ercot.com/gridinfo/generation).

REFERENCES

Adams-Selin, R. D., and C. L. Ziegler, 2016: Forecasting hail using a one-dimensional hail growth model within WRF. *Mon. Wea. Rev.*, **144**, 4919–4939, https://doi.org/10.1175/MWR-D-16-0027.1.

Alpert, J. C., and V. K. Kumar, 2007: Radial wind super-obs from the WSR-88D radars in the NCEP operational assimilation system. *Mon. Wea. Rev.*, **135**, 1090–1109, https://doi.org/10.1175/MWR3324.1.

American Wind Energy Association, 2019: U.S. wind industry third quarter 2019 market report. AWEA Rep., 47 pp., https://www.awea.org/resources/publications-and-reports/market-reports/2019-u-s-wind-industry-market-reports/3q2019_marketreport-(1).

——, 2020: Wind powers America annual report 2019. AWEA Rep., https://www.awea.org/resources/publications-and-reports/market-reports/2019-u-s-wind-industry-market-reports.

Appleby-Thomas, G. J., P. J. Hazell, and G. Dahini, 2011: On the response of two commercially-important CFRP structures to multiple ice impacts. *Compos. Struct.*, **93**, 2619–2627, https://doi.org/10.1016/j.compstruct.2011.04.029.

Bartolomé, L., and J. Teuwen, 2019: Prospective challenges in the experimentation of the rain erosion on the leading edge of wind turbine blades. *Wind Energy*, **22**, 140–151, https://doi.org/10.1002/we.2272.

Bech, J. I., C. B. Hasager, and C. Bak, 2018: Extending the life of wind turbine blade leading edges by reducing the tip speed during extreme precipitation events. *Wind Energy Sci.*, **3**, 729–748, https://doi.org/10.5194/wes-3-729-2018.

Brown, T. M., W. H. Pogorzelski, and I. M. Giammanco, 2015: Evaluating hail damage using property insurance claims data. *Wea. Climate Soc.*, **7**, 197–210, https://doi.org/10.1175/WCAS-D-15-0011.1.

Buechler, D. E., and S. J. Goodman, 1990: Echo size and asymmetry: Impact on NEXRAD storm identification. *J. Appl. Meteor.*, **29**, 962–969, https://doi.org/10.1175/1520-0450(1990)029<0962:ESAAIO>2.0.CO;2.

Carroll, J., A. McDonald, and D. McMillan, 2016: Failure rate, repair time and unscheduled O&M cost analysis of offshore wind turbines. *Wind Energy*, **19**, 1107–1119, https://doi.org/10.1002/we.1887.

Carvalho, D., A. Rocha, M. Gómez-Gesteira, and C. Santos, 2012: A sensitivity study of the WRF Model in wind simulation for an area of high wind energy. *Environ. Modell. Software*, **33**, 23–34, https://doi.org/10.1016/j.envsoft.2012.01.019.

Chen, F., and J. Dudhia, 2001: Coupling an advanced land surface–hydrology model with the Penn State–NCAR MM5 modeling system. Part I: Model implementation and sensitivity. *Mon. Wea. Rev.*, **129**, 569–585, https://doi.org/10.1175/1520-0493(2001)129<0569:CAALSH>2.0.CO;2.

Cintineo, J. L., T. M. Smith, V. Lakshmanan, H. E. Brooks, and K. L. Ortega, 2012: An objective high-resolution hail climatology of the contiguous United States. *Wea. Forecasting*, **27**, 1235–1248, https://doi.org/10.1175/WAF-D-11-00151.1.

——, M. J. Pavolonis, J. M. Sieglaff, and D. T. Lindsey, 2014: An empirical model for assessing the severe weather potential of developing convection. *Wea. Forecasting*, **29**, 639–653, https://doi.org/10.1175/WAF-D-13-00113.1.

Cortés, E., F. Sánchez, A. O'Carroll, B. Madramany, M. Hardiman, and T. M. Young, 2017: On the material characterisation of wind turbine blade coatings: The effect of interphase coating–laminate adhesion on rain erosion performance. *Materials*, **10**, 1146, https://doi.org/10.3390/ma10101146.

Crum, T. D., R. E. Saffle, and J. W. Wilson, 1998: An update on the NEXRAD program and future WSR-88D support to operations. *Wea. Forecasting*, **13**, 253–262, https://doi.org/10.1175/1520-0434(1998)013<0253:AUOTNP>2.0.CO;2.

Dee, D. P., and Coauthors, 2011: The ERA-Interim reanalysis: Configuration and performance of the data assimilation system. *Quart. J. Roy. Meteor. Soc.*, **137**, 553–597, https://doi.org/10.1002/qj.828.

Diffendorfer, J. E., R. Compton, L. Kramer, Z. Ancona, and D. Norton, 2017: Onshore industrial wind turbine locations for the United States, version 1.2. USGS Publ., 5 pp., https://pubs.er.usgs.gov/publication/ds817.

Dudhia, J., 1989: Numerical study of convection observed during the winter monsoon experiment using a mesoscale two-dimensional model. *J. Atmos. Sci.*, **46**, 3077–3107, https://doi.org/10.1175/1520-0469(1989)046<3077:NSOCOD>2.0.CO;2.

Dykes, K., and Coauthors, 2014: Effect of tip-speed constraints on the optimized design of a wind turbine. National Renewable Energy Laboratory Doc. NREL/TP-5000-61726, 65 pp., https://www.nrel.gov/docs/fy15osti/61726.pdf.

Eisenberg, D., S. Laustsen, and J. Stege, 2018: Wind turbine blade coating leading edge rain erosion model: Development and validation. *Wind Energy*, **21**, 942–951, https://doi.org/10.1002/we.2200.

Fan, J., and Coauthors, 2017: Cloud-resolving model intercomparison of an MC3E squall line case: Part I—Convective updrafts. *J. Geophys. Res. Atmos.*, **122**, 9351–9378, https://doi.org/10.1002/2017JD026622.

Feng, Z., L. R. Leung, R. A. Houze Jr., S. Hagos, J. Hardin, Q. Yang, B. Han, and J. Fan, 2018: Structure and evolution of mesoscale convective systems: Sensitivity to cloud microphysics in convection-permitting simulations over the United States. *J. Adv. Model. Earth Syst.*, **10**, 1470–1494, https://doi.org/10.1029/2018MS001305.

Fertig, E., 2019: Simulating subhourly variability of wind power output. *Wind Energy*, **22**, 1275–1287, https://doi.org/10.1002/we.2354.

Fiore, G., G. E. Camarinha Fujiwara, and M. S. Selig, 2015: A damage assessment for wind turbine blades from heavy atmospheric particles. *53rd AIAA Aerospace Sciences Meeting*, Kissimmee, FL, AIAA, 2015-1495, https://m-selig.ae.illinois.edu/pubs/FioreFujiwaraSelig-2015-AIAA-2015-1495.pdf.

Fitch, A. C., 2015: Climate impacts of large-scale wind farms as parameterized in a global climate model. *J. Climate*, **28**, 6160–6180, https://doi.org/10.1175/JCLI-D-14-00245.1.

——, J. B. Olson, J. K. Lundquist, J. Dudhia, A. K. Gupta, J. Michalakes, and I. Barstad, 2012: Local and mesoscale impacts of wind farms as parameterized in a mesoscale NWP model. *Mon. Wea. Rev.*, **140**, 3017–3038, https://doi.org/10.1175/MWR-D-11-00352.1.

——, ——, and ——, 2013: Parameterization of wind farms in climate models. *J. Climate*, **26**, 6439–6458, https://doi.org/10.1175/JCLI-D-12-00376.1.

Forget, P., M. Saillard, C.-A. Guérin, J. Testud, and E. Le Bouar, 2016: On the use of x-band weather radar for wind field retrieval in coastal zone. *J. Atmos. Oceanic Technol.*, **33**, 899–917, https://doi.org/10.1175/JTECH-D-15-0206.1.

Grabowski, W. W., H. Morrison, S.-I. Shima, G. C. Abade, P. Dziekan, and H. Pawlowska, 2019: Modeling of cloud

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

microphysics: Can we do better? *Bull. Amer. Meteor. Soc.*, **100**, 655–672, https://doi.org/10.1175/BAMS-D-18-0005.1.

Hahmann, A. N., and Coauthors, 2020: The making of the New European Wind Atlas, part 1: Model sensitivity. *Geosci. Model Dev.*, https://doi.org/10.5194/gmd-2019-349, in press.

Hasager, C., F. Vejen, J. Bech, W. Skrzypiński, A.-M. Tilg, and M. Nielsen, 2020: Assessment of the rain and wind climate with focus on wind turbine blade leading edge erosion rate and expected lifetime in Danish seas. *Renewable Energy*, **149**, 91–102, https://doi.org/10.1016/j.renene.2019.12.043.

Hawbecker, P., S. Basu, and L. Manuel, 2017: Realistic simulations of the July 1, 2011 severe wind event over the Buffalo Ridge Wind Farm. *Wind Energy*, **20**, 1803–1822, https://doi.org/10.1002/we.2122.

Herring, R., K. Dyer, F. Martin, and C. Ward, 2019: The increasing importance of leading edge erosion and a review of existing protection solutions. *Renewable Sustainable Energy Rev.*, **115**, 109382, https://doi.org/10.1016/j.rser.2019.109382.

International Energy Agency, 2020: Erosion of wind turbine blades. *Topical Expert Meeting 98*, Roskilde, Denmark, Implementing Agreement for Co-operation in the Research, Development and Deployment of Wind Turbine Systems Task 11, https://community.ieawind.org/task11/ourlibrary.

Jiménez, P. A., J. Dudhia, J. F. González-Rouco, J. Navarro, J. P. Montávez, and E. García-Bustamante, 2012: A revised scheme for the WRF surface layer formulation. *Mon. Wea. Rev.*, **140**, 898–918, https://doi.org/10.1175/MWR-D-11-00056.1.

Johnson, M., Y. Jung, D. T. Dawson, and M. Xue, 2016: Comparison of simulated polarimetric signatures in idealized supercell storms using two-moment bulk microphysics schemes in WRF. *Mon. Wea. Rev.*, **144**, 971–996, https://doi.org/10.1175/MWR-D-15-0233.1.

Jonkman, J., S. Butterfield, W. Musial, and G. Scott, 2009: Definition of a 5-MW reference wind turbine for offshore system development. NREL Tech. Rep. NREL/TP-500-38060, 75 pp., https://www.nrel.gov/docs/fy09osti/38060.pdf.

Kain, J. S., and J. M. Fritsch, 1993: Convective parameterization for mesoscale models: The Kain–Fritsch scheme. *The Representation of Cumulus Convection in Numerical Models*, *Meteor. Monogr.*, No. 24, Amer. Meteor. Soc., 165–170.

——, S. J. Weiss, J. J. Levit, M. E. Baldwin, and D. R. Bright, 2006: Examination of convection-allowing configurations of the WRF Model for the prediction of severe convective weather: The SPC/NSSL Spring Program 2004. *Wea. Forecasting*, **21**, 167–181, https://doi.org/10.1175/WAF906.1.

Kim, H., and K. T. Kedward, 2000: Modeling hail ice impacts and predicting impact damage initiation in composite structures. *AIAA J.*, **38**, 1278–1288, https://doi.org/10.2514/3.14545.

Letson, F., T. J. Shepherd, R. J. Barthelmie, and S. C. Pryor, 2020a: Modelling hail and convective storms with WRF for wind energy applications. *J. Phys. Conf. Ser.*, **1452**, 012051, https://doi.org/10.1088/1742-6596/1452/1/012051.

——, R. J. Barthelmie, and S. C. Pryor, 2020b: Radar-derived precipitation climatology for wind turbine blade leading edge erosion. *Wind Energy Sci.*, **5**, 331–347, https://doi.org/10.5194/wes-5-331-2020.

Loomis, I., 2018: Hail causes the most storm damage costs across North America. *Eos, Trans. Amer. Geophys. Union*, **99**, 3–4, https://doi.org/10.1029/2018EO104487.

Macdonald, H., D. Infield, D. H. Nash, and M. M. Stack, 2016: Mapping hail meteorological observations for prediction of erosion in wind turbines. *Wind Energy*, **19**, 777–784, https://doi.org/10.1002/we.1854.

Masters, F. J., P. J. Vickery, P. Bacon, and E. N. Rappaport, 2010: Toward objective, standardized intensity estimates from surface wind speed observations. *Bull. Amer. Meteor. Soc.*, **91**, 1665–1682, https://doi.org/10.1175/2010BAMS2942.1.

Milbrandt, J., and M. Yau, 2005: A multimoment bulk microphysics parameterization. Part I: Analysis of the role of the spectral shape parameter. *J. Atmos. Sci.*, **62**, 3051–3064, https://doi.org/10.1175/JAS3534.1.

Mishnaevsky, L., Jr., and J. Sütterlin, 2019: Micromechanical model of surface erosion of polyurethane coatings on wind turbine blades. *Polym. Degrad. Stab.*, **166**, 283–289, https://doi.org/10.1016/j.polymdegradstab.2019.06.009.

——, and K. Thomsen, 2020: Costs of repair of wind turbine blades: Influence of technology aspects. *Wind Energy*, https://doi.org/10.1002/we.2552, in press.

——, K. Branner, H. N. Petersen, J. Beauson, M. McGugan, and B. F. Sørensen, 2017: Materials for wind turbine blades: An overview. *Materials*, **10**, 1285, https://doi.org/10.3390/ma10111285.

Mlawer, E. J., S. J. Taubman, P. D. Brown, M. J. Iacono, and S. A. Clough, 1997: Radiative transfer for inhomogeneous atmospheres: RRTM, a validated correlated-*k* model for the longwave. *J. Geophys. Res.*, **102**, 16 663–16 682, https://doi.org/10.1029/97JD00237.

Morrison, H., and J. Milbrandt, 2011: Comparison of two-moment bulk microphysics schemes in idealized supercell thunderstorm simulations. *Mon. Wea. Rev.*, **139**, 1103–1130, https://doi.org/10.1175/2010MWR3433.1.

Nakanishi, M., and H. Niino, 2006: An improved Mellor–Yamada level-3 model: Its numerical stability and application to a regional prediction of advection fog. *Bound.-Layer Meteor.*, **119**, 397–407, https://doi.org/10.1007/s10546-005-9030-8.

Nielsen, J. S., D. Tcherniak, and M. D. Ulriksen, 2020: A case study on risk-based maintenance of wind turbine blades with structural health monitoring. *Struct. Infrastruct. Eng.*, https://doi.org/10.1080/15732479.2020.1743326, in press.

Ning, A., and K. Dykes, 2014: Understanding the benefits and limitations of increasing maximum rotor tip speed for utility-scale wind turbines. *J. Phys. Conf. Ser.*, **524**, 012087, https://doi.org/10.1088/1742-6596/524/1/012087.

NOAA, 2004: Automated Surface Observing System (ASOS) release note, software version 2.79. NWS Doc., 10 pp., https://www.weather.gov/media/asos/ASOS%20Implementation/release_notes.279_final.pdf.

——, 2019: Storm events details, 2014. Storm Events Database, accessed 10 January 2020, https://www.ncdc.noaa.gov/stormevents/.

Office of the Federal Coordinator for Meteorology, 2016: System concepts, responsibilities, and procedures. Part A, Doppler Radar Meteorological Observations (WSR-88D). NOAA Federal Meteorological Handbook 11: FCM-H11A-2016, 25 pp., https://www.ofcm.gov/publications/fmh/FMH11/2016FMH11PTA.pdf.

——, 2017: Products and algorithms. Part C, Doppler Radar Meteorological Observations (WSR-88D). NOAA Federal Meteorological Handbook 11: FCM-H11C-2017, 394 pp., https://www.ofcm.gov/publications/fmh/FMH11/fmh11partC.pdf.

Papaelias, M., and F. P. G. Márquez, 2020: Wind turbine inspection and condition monitoring. *Non-Destructive Testing and Condition Monitoring Techniques for Renewable Energy Industrial Assets*, A. Karyotakis and M. Papaelias, Eds., Elsevier, 19–29.

Prein, A. F., and G. J. Holland, 2018: Global estimates of damaging hail hazard. *Wea. Climate Extremes*, **22**, 10–23, https://doi.org/10.1016/j.wace.2018.10.004.

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

——, and Coauthors, 2015: A review on regional convection-permitting climate modeling: Demonstrations, prospects, and challenges. *Rev. Geophys.*, **53**, 323–361, https://doi.org/10.1002/2014RG000475.

——, C. Liu, K. Ikeda, S. B. Trier, R. M. Rasmussen, G. J. Holland, and M. P. Clark, 2017: Increased rainfall volume from future convective storms in the US. *Nat. Climate Change*, **7**, 880–884, https://doi.org/10.1038/s41558-017-0007-7.

Prósper, M. A., C. Otero-Casal, F. C. Fernández, and G. Miguez-Macho, 2019: Wind power forecasting for a real onshore wind farm on complex terrain using WRF high resolution simulations. *Renewable Energy*, **135**, 674–686, https://doi.org/10.1016/j.renene.2018.12.047.

Pryor, S. C., and A. N. Hahmann, 2019: Downscaling wind. *Oxford Research Encyclopedia, Climate Science*, H. von Storch, Ed., Oxford University Press, https://doi.org/10.1093/acrefore/9780190228620.013.730.

——, R. J. Barthelmie, and T. Shepherd, 2018: The influence of real-world wind turbine deployments on regional climate. *J. Geophys. Res. Atmos.*, **123**, 5804–5826, https://doi.org/10.1029/2017JD028114.

——, ——, and ——, 2020a: 20% of US electricity from wind will have limited impacts on system efficiency and regional climate. *Sci. Rep.*, **10**, 541, https://doi.org/10.1038/s41598-019-57371-1.

——, T. J. Shepherd, P. J. H. Volker, A. N. Hahmann, and R. J. Barthelmie, 2020b: "Wind theft" from onshore wind turbine arrays: Sensitivity to wind farm parameterization and resolution. *J. Appl. Meteor. Climatol.*, **59**, 153–174, https://doi.org/10.1175/JAMC-D-19-0235.1.

——, ——, ——, ——, and ——, 2020c: Diagnosing systematic differences in predicted wind turbine array-array interactions. *J. Phys. Conf. Ser.*, **1618**, 062023, https://doi.org/10.1088/1742-6596/1618/6/062023.

Púčik, T., and Coauthors, 2017: Future changes in European severe convection environments in a regional climate model ensemble. *J. Climate*, **30**, 6771–6794, https://doi.org/10.1175/JCLI-D-16-0777.1.

Punge, H., and M. Kunz, 2016: Hail observations and hailstorm characteristics in Europe: A review. *Atmos. Res.*, **176–177**, 159–184, https://doi.org/10.1016/j.atmosres.2016.02.012.

Rand, J. T., L. A. Kramer, C. P. Garrity, B. D. Hoen, J. E. Diffendorfer, H. E. Hunt, and M. Spears, 2020: A continuously updated, geospatially rectified database of utility-scale wind turbines in the United States. *Sci. Data*, **7**, 15, https://doi.org/10.1038/s41597-020-0353-6.

Reynolds, R. W., and D. B. Chelton, 2010: Comparisons of daily sea surface temperature analyses for 2007–08. *J. Climate*, **23**, 3545–3562, https://doi.org/10.1175/2010JCLI3294.1.

Scaff, L., A. F. Prein, Y. Li, C. Liu, R. Rasmussen, and K. Ikeda, 2020: Simulating the convective precipitation diurnal cycle in North America's current and future climate. *Climate Dyn.*, **55**, 369–382, https://doi.org/10.1007/s00382-019-04754-9.

Schmitt, I. V., and V. Chester, 2009: A quality control algorithm for the ASOS ice free wind sensor. *13th Conf. on Integrated Observing and Assimilation Systems for Atmosphere, Oceans, and Land Surface*, Phoenix, AZ, Amer. Meteor. Soc., 12A.3, https://ams.confex.com/ams/pdfpapers/145755.pdf.

Seo, B.-C., B. Dolan, W. F. Krajewski, S. A. Rutledge, and W. Petersen, 2015: Comparison of single-and dual-polarization–based rainfall estimates using NEXRAD data for the NASA Iowa Flood Studies project. *J. Hydrometeor.*, **16**, 1658–1675, https://doi.org/10.1175/JHM-D-14-0169.1.

Shepherd, T. J., R. J. Barthelmie, and S. C. Pryor, 2020: Sensitivity of wind turbine array downstream effects to the parameterization used in WRF. *J. Appl. Meteor. Climatol.*, **59**, 333–361, https://doi.org/10.1175/JAMC-D-19-0135.1.

Shpund, J., and Coauthors, 2019: Simulating a mesoscale convective system using WRF with a new spectral bin microphysics: 1: Hail vs graupel. *J. Geophys. Res. Atmos.*, **124**, 14 072–14 101, https://doi.org/10.1029/2019JD030576.

Skamarock, W. C., and Coauthors, 2008: A description of the Advanced Research WRF version 3. NCAR Tech. Note NCAR/TN-475+STR, 113 pp., https://doi.org/10.5065/D68S4MVH.

Snook, N., Y. Jung, J. Brotzge, B. Putnam, and M. Xue, 2016: Prediction and ensemble forecast verification of hail in the supercell storms of 20 May 2013. *Wea. Forecasting*, **31**, 811–825, https://doi.org/10.1175/WAF-D-15-0152.1.

Stephenson, D. B., 2000: Use of the "odds ratio" for diagnosing forecast skill. *Wea. Forecasting*, **15**, 221–232, https://doi.org/10.1175/1520-0434(2000)015<0221:UOTORF>2.0.CO;2.

Stull, R. B., 2017: *Practical Meteorology: An Algebra-Based Survey of Atmospheric Science*. AVP International, University of British Columbia, 940 pp.

Tang, B. H., V. A. Gensini, and C. R. Homeyer, 2019: Trends in United States large hail environments and observations. *npj Climate Atmos. Sci.*, **2**, 45, https://doi.org/10.1038/S41612-019-0103-7.

Tao, W. K., D. Wu, S. Lang, J. D. Chern, C. Peters-Lidard, A. Fridlind, and T. Matsui, 2016: High-resolution NU-WRF simulations of a deep convective-precipitation system during MC3E: Further improvements and comparisons between Goddard microphysics schemes and observations. *J. Geophys. Res. Atmos.*, **121**, 1278–1305, https://doi.org/10.1002/2015JD023986.

Tilg, A. M., C. B. Hasager, H. J. Kirtzel, and P. Hummelshoj, 2020: Brief communication: Nowcasting of precipitation for leading edge erosion-safe mode. *Wind Energy Sci.*, **5**, 977–981, https://doi.org/10.5194/wes-5-977-2020.

Tomaszewski, J. M., and J. K. Lundquist, 2020: Simulated wind farm wake sensitivity to configuration choices in the Weather Research and Forecasting Model version 3.8.1. *Geosci. Model Dev.*, **13**, 2645–2662, https://doi.org/10.5194/gmd-13-2645-2020.

Traphan, D., I. Herráez, P. Meinlschmidt, F. Schlüter, J. Peinke, and G. Gülker, 2018: Remote surface damage detection on rotor blades of operating wind turbines by means of infrared thermography. *Wind Energy Science*, **3**, 639–650, https://doi.org/10.5194/wes-3-639-2018.

Wallace, R., K. Friedrich, E. A. Kalina, and P. Schlatter, 2019: Using operational radar to identify deep hail accumulations from thunderstorms. *Wea. Forecasting*, **34**, 133–150, https://doi.org/10.1175/WAF-D-18-0053.1.

Weisman, M. L., W. C. Skamarock, and J. B. Klemp, 1997: The resolution dependence of explicitly modeled convective systems. *Mon. Wea. Rev.*, **125**, 527–548, https://doi.org/10.1175/1520-0493(1997)125<0527:TRDOEM>2.0.CO;2.

Wendt, N. A., I. L. Jirak, and C. J. Melick, 2016: Verification of severe weather proxies from the NSSL-WRF for hail forecasting. *28th Conf. on Severe Local Storms*, Portland, OR, Amer. Meteor. Soc., 110, https://ams.confex.com/ams/28SLS/webprogram/Manuscript/Paper300913/nawendt_sls28_ext_abstract.pdf.

Wetzel, K., 2019: How do we determine optimal inspection frequency? The case for routine blade inspections. *Windtech*

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC

International, 20–21, https://www.windtech-international.com/editorial-features/the-case-for-routine-blade-inspections.

Wilks, D. S., 2011: *Statistical Methods in the Atmospheric Sciences*. 3rd ed. Elsevier, 676 pp.

Wiser, R. H., and M. Bolinger, 2019: Benchmarking anticipated wind project lifetimes: Results from a survey of U.S. wind industry professionals. Berkeley Lab: Electricity Markets and Policy Group, https://emp.lbl.gov/publications/benchmarking-anticipated-wind-project.

Witt, A., M. D. Eilts, G. J. Stumpf, J. Johnson, E. D. W. Mitchell, and K. W. Thomas, 1998: An enhanced hail detection algorithm for the WSR-88D. *Wea. Forecasting*, **13**, 286–303, https://doi.org/10.1175/1520-0434(1998)013<0286:AEHDAF>2.0.CO;2.

Yang, Q., R. A. Houze, L. R. Leung, and Z. Feng, 2017: Environments of long-lived mesoscale convective systems over the central United States in convection permitting climate simulations. *J. Geophys. Res. Atmos.*, **122**, 13 288–13 307, https://doi.org/10.1002/2017JD027033.

Zhang, S., K. Dam-Johansen, S. Nørkjær, P. L. Bernad Jr., and S. Kiil, 2015: Erosion of wind turbine blade coatings—Design and analysis of jet-based laboratory equipment for performance evaluation. *Prog. Org. Coat.*, **78**, 103–115, https://doi.org/10.1016/j.porgcoat.2014.09.016.

Unauthenticated | Downloaded 10/06/25 12:36 PM UTC